

Single Systems

Qiskit Implementation Vectors & Matrices

$$\text{ket0} = |0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \text{ket1} = |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$M1 = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \quad M2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$M = \frac{1}{2}M1 + \frac{1}{2}M2 = \begin{bmatrix} 1/2 & 1/2 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 1/2 & 0 \\ 0 & 1/2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1/2 \\ 0 & 1/2 \end{bmatrix} = \begin{bmatrix} 1 & 0.5 \\ 0 & 0.5 \end{bmatrix}$$

$$M1|1\rangle = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0+1 \\ 0+0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$M1M2 = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1+0 & 0+1 \\ 0+0 & 0+0 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$$

$$MM = \begin{bmatrix} 1 & 1/2 \\ 0 & 1/2 \end{bmatrix} \begin{bmatrix} 1 & 1/2 \\ 0 & 1/2 \end{bmatrix} = \begin{bmatrix} 1+0 & 1/2+1/4 \\ 0+0 & 0+1/4 \end{bmatrix} = \begin{bmatrix} 1 & 3/4 \\ 0 & 1/4 \end{bmatrix} = \begin{bmatrix} 1 & 0.75 \\ 0 & 0.25 \end{bmatrix}$$

State, measurements & operations

$$u = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} \quad v = \begin{bmatrix} \frac{1+2i}{3} \\ -2/3 \end{bmatrix} \quad w = \begin{bmatrix} 1/3 \\ 2/3 \end{bmatrix}$$